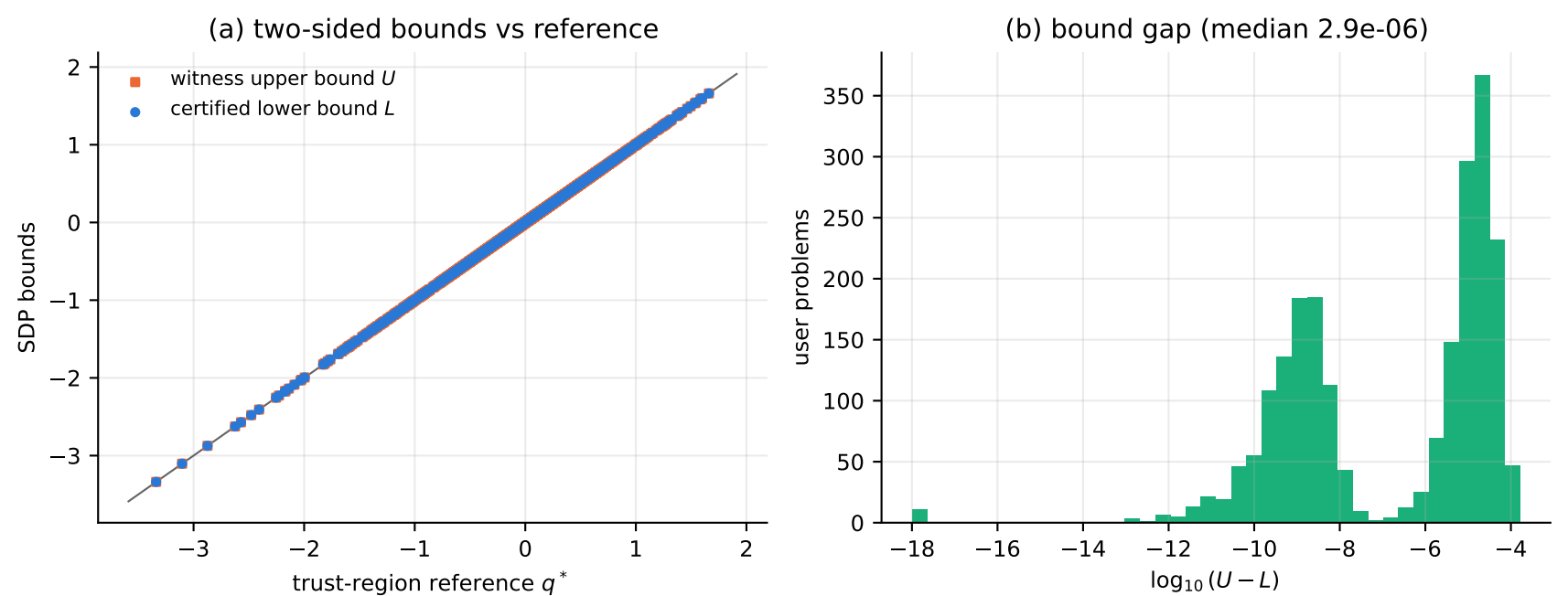

(a) two-sided bounds vs reference
witness upper bound U
certified lower bound L
SDP bounds
trust-region reference q^*
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
(b) bound gap (median 2.9e-06)
user problems
$\log_{10}(U-L)$
350
300
250
200
150
100
50
0
−18
−16
−14
−12
−10
−8
−6
−4